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Begonnen am Thursday, 11. May 2023, 17:09

Status Beendet

Beendet am Thursday, 11. May 2023, 17:20

Verbrauchte Zeit 10 Minuten 20 Sekunden

Bewertung 4,47 von 10,00 (44,67%)

Frage 1

Vollständig

Erreichte Punkte 1,00 von 1,00

According to R. Sutton and Minsky, what is the problem of credit-assignment?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Assigning credit or blame for overall outcomes to each of the decisions that contributed to those outcomes
- ☐ b. Identifying models that get high rewards
- ☐ c. Identifying actions that led to high rewards
- ☐ d. Assigning credit or blame for overall episodes that contributed to the expectation of the return.

Frage 2

Vollständig

Erreichte Punkte 0,00 von 1,00

Which of the following sentences are true

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Variance in TD(0) increases with the delay of the reward
- ☐ b. Variance in MC increases with the delay of the reward
- ☒ c. $TD(\lambda)$ assigns credit faster than TD(0)
- ☒ d. TD methods corrects the bias faster than MC

Frage 3

Vollständig

Erreichte Punkte 1,00 von 1,00

The on-site variance ω is defined as:

$$\omega(s, a) = \text{Var}_r[r \mid s, a] + \text{Var}_{s', a'}[q^\pi(s', a') \mid s, a]$$

In which of the following situations, this on-site variance is zero?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. For TD(0) and deterministic rewards
- ☒ b. For deterministic state transition environments, deterministic rewards and deterministic policies.
- ☐ c. For non-delayed rewards environments
- ☐ d. For deterministic state transition environments

Frage 4

Vollständig

Erreichte Punkte 0,00 von 1,00

In MC, and under certain assumptions, the number of states affected by the on-site variance is:

$$a_k = \sum_{t=0}^k \left(1 - \left(1 - \frac{c_t}{N_{t-1}}\right)^{n_t}\right) N_{t-1}$$

where N_t is the number of states reachable after t steps, n_t is the number of states affected by the on-site variance at time t and c_t is the random average connectivity of s_t to s_{t-1} .

What of the following sentences concerning k can be deduced from the expression above? (collorary of the theorem)

Wählen Sie eine oder mehrere Antworten:

- ☐ a. For **small** k , and after some time $t > \hat{t}$, the number of states affected by the on-site variance at step k **grows linearly** with k
- ☐ b. For **large** k , and after some time $t > \hat{t}$, the number of states affected by the on-site variance at step k **grows linearly** with k
- ☐ c. For **small** k , the number of states affected by the on-site variance at step k **grows exponentially** with k
- ☒ d. For **large** k , the number of states affected by the on-site variance at step k **grows exponentially** with k

Frage 5

Vollständig

Erreichte Punkte 0,67 von 1,00

If we make the expected future rewards equals to zero (i.e. via reward redistribution), the Q -values are just the average of the immediate reward. However, why this is not possible in an MDP setting?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Because in this case (future expected reward equal to zero), the reward is obtained before we actually get in the environment, and therefore, is not causal and breaks the Markov property.
- ☐ b. Because in order to have future expected reward to zero, states have to be enriched with the past, and therefore, breaks the Markov assumption
- ☒ c. Because in this case (future expected reward equal to zero), the reward depends also on the previous state and action, and therefore, is not Markov.
- ☐ d. It is completely possible in an MDP setting.

Frage 6

Vollständig

Erreichte Punkte 0,00 von 1,00

What is the definition of a sequence-Markov decision process (SDP), as seen in class?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** and has **Markov transition probabilities** but a **reward that is not required to be Markov**
- ☒ b. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov transition probabilities** but a **policy that is not required to be Markov**
- ☐ c. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov reward** and has **Markov policy** but a **transition probabilities that is not required to be Markov**
- ☐ d. A sequence-Markov decision process (SDP) is defined as a decision process which is equipped with a **Markov policy** but a **transition probabilities and reward that is not required to be Markov**

Frage 7

Vollständig

Erreichte Punkte 0,00 von 1,00

What is the definition of **return-equivalent** SDPs?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same return per episode**
- ☒ b. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return per episode**
- ☐ c. Two SDPs are return equivalent if they differ only in their reward distributions but have the **same expected return**

Frage 8

Vollständig

Erreichte Punkte 0,00 von 1,00

What is the **explaining away problem** in MDPs, when a function g is used to predict the return based on the sequence of actions and states?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Since later states are used for return prediction, earlier actions causing reward are missed.
- ☒ b. Previous actions can explain away future states, breaking the Markov property
- ☐ c. In an MDP, one only need the difference of states actions to make a prediction and not the whole sequence.
- ☐ d. Since the sequence of states and actions are mutually independent, the prediction needs every component to explain the return.

Frage 9

Vollständig

Erreichte Punkte 0,80 von 1,00

What of the following statements are true when referring to differences between LSTM and Feed Forward networks (FFN) for making future predictions?

Wählen Sie eine oder mehrere Antworten:

- ☒ a. FFNs are prone to prediction errors since they have to extract information from every state to make a prediction in every timestep
- ☐ b. LSTM will only learn to store information if a state-action pair has a strong evidence for a change in the expected return.
- ☒ c. FFNs are not suited for processing sequences.
- ☒ d. LSTM tends to have smaller prediction errors since reuses past information
- ☒ e. Prediction errors of LSTMs are much more likely to cancel via prediction differences. Since consecutive predictions of LSTMs rely on the same internal states, they usually have highly correlated errors.

Frage 10

Vollständig

Erreichte Punkte 1,00 von 1,00

What is the main difference between sensitivity analysis (SA) and contribution analysis (CA)?

Wählen Sie eine oder mehrere Antworten:

- ☐ a. **SA** is gradient-based, while **CA** is not
- ☐ b. **CA** is gradient-based, while **SA** is not
- ☒ c. **SA** focuses on how a small change in the input change the output, and **CA** focuses on how the input is responsible of the output
- ☐ d. **CA** focuses on how a small change in the input change the output, and **SA** focuses on how the input is responsible of the output

◀ [Exam] Connections between control and inference

Direkt zu:

[Exam] PG 1 ►